

UNIVERSITY OF MORATUWA

Faculty of Engineering



EE4410 – Electrical Services for Buildings

ASSIGNMENT 03 HEAT LOAD CALCULATION

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1.0 Introduction

Following the proposal of an energy-efficient LED lighting arrangement for the Electrical Engineering Computer Laboratory located on the ground floor of the Sumanadasa Building, this report presents a comprehensive heat load calculation for the facility. Building upon the parameters of the newly proposed lighting system, this study evaluates all internal and external heat sources to determine the total required cooling capacity. Based on these technical calculations, a suitable, energy-efficient air conditioning system is proposed and justified to ensure optimal thermal comfort for the space.



Figure 01 – Laboratory (DIALux view)

2.0 Room Details

2.1 Room Dimensions

To establish the thermal boundaries for the heat load calculation, the geometric dimensions of the Computer Laboratory were identified using the official floor plan, as outlined below.

Dimension	Value
Length	16.942m
Width	9.449m
Hight	3.5m
Wall thickness	0.16m

2.2 Doors and Windows

The computer laboratory features multiple types of doors and windows. Accurately documenting their dimensions and materials is crucial, as they represent significant pathways for heat transfer and are essential parameters in the overall heat load calculation.

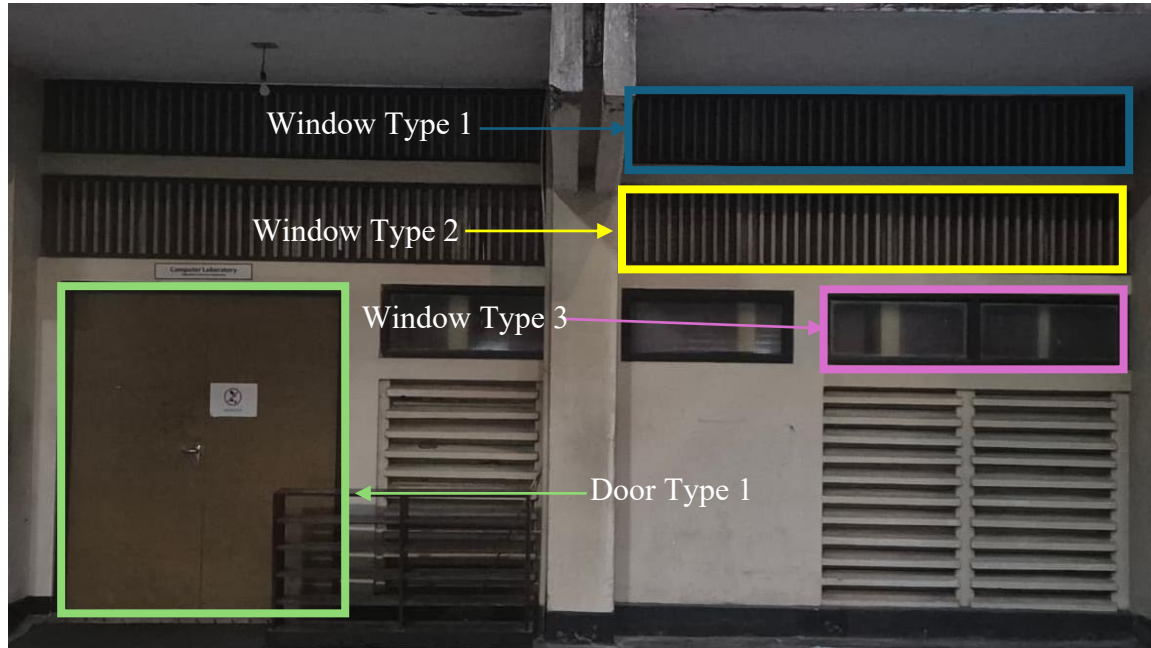


Figure 02 – Front view of the laboratory

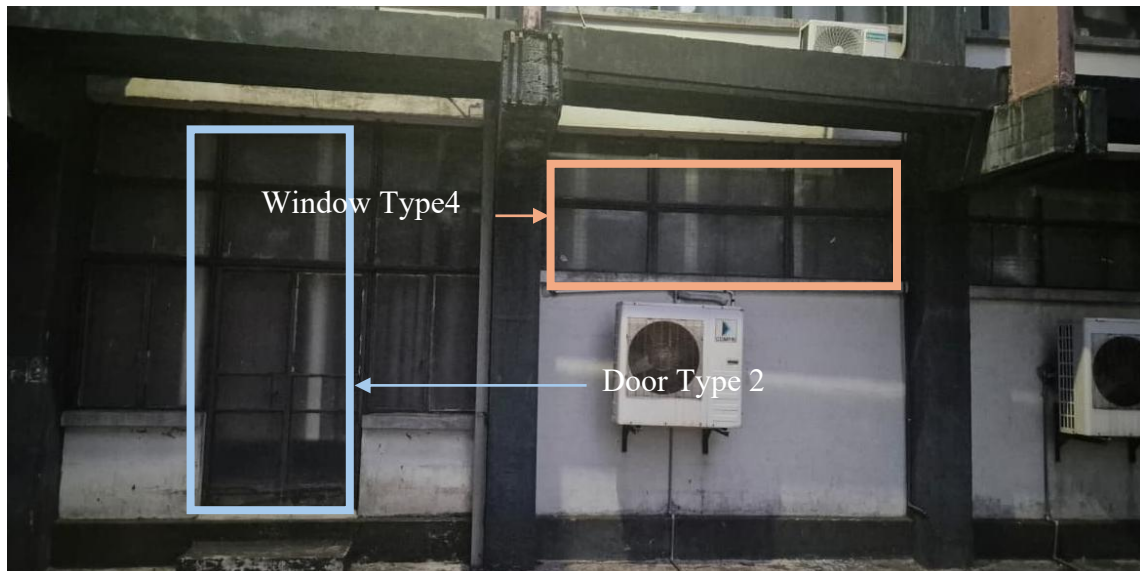


Figure 03 – Rear view of the laboratory

- **Doors**

There are three different doors located within the laboratory. Only one acts as the primary access point, while the other two remain permanently closed. The specific dimensions and characteristics of each door are summarized in the table below.

Door	Location	Characteristics	Dimensions
Door Type 1	Front side right corner	• Made of wood. • Two door arrangement.	• Width – 1.67m • Hight – 2.01m
Door Type 2	Front side left corner	• Made of wood. • Single door arrangement.	• Width – 1.07m • Hight – 2.01m
Door Type 3	Rear side left corner	• Made of glass. • Single door arrangement.	• Width – 1.07m • Hight – 2.01m

- **Windows**

The laboratory features four distinct types of windows as indicated in figure 2 and 3. The physical dimensions of these windows are detailed in the following table.

Window	Number of Windows	Dimensions
Window Type 1	3	• Width – 3.25m • Hight – 0.47m
Window Type 2	5	• Width – 3.25m • Hight – 0.53m
Window Type 3	10	• Width – 1m • Hight – 0.35m
Window Type 4	4	• Width – 3.2m • Hight – 1.24m

2.3 Orientation of the Laboratory

The laboratory is oriented along a North-South axis, with Door Types 1 and 2 and Window Types 1, 2, and 3 facing South, and Door Type 3 and Window Type 4 facing North. Given this orientation and the presence of architectural shading canopies, the exposure to direct sunlight and the resulting solar heat gain is treated as negligible.



Figure 04 – Orientation of the laboratory

03. Heat Sources

To accurately size the proposed air conditioning system, it is essential to identify and quantify all thermal loads contributing to the temperature rise within the Computer Laboratory. A comprehensive thermal analysis requires evaluating every element that introduces sensible or latent heat into the space. For the purpose of this calculation, these thermal contributors are classified into two primary categories as internal heat sources and external heat sources.

03.1 Internal Heat Sources

Internal heat gains account for the sensible and latent thermal energy emitted by occupants, equipment, and lighting within the facility. The following internal heat sources have been identified for this specific laboratory environment.

Internal Heat Source	Details
Desktop Computers	• 74 desktop computers available. • Assume one monitor is 50W. • Assume desktop computer specifications are 2.4GHz processor, 8GB RAM
Students + Staff	• Seating is available for 75 people at the laboratory. • Assume 60 people are present at a given peak time. • Laboratory is assumed as a moderately active office work area.
Lighting	• A total of 22 bulbs were proposed in the lighting redesign. • Proposed single bulb is 34W. • Utilization factor for the lights is assumed as 0.9.

03.2 External Heat Sources

External heat gains represent the sensible thermal loads transmitted into the laboratory through its structural boundaries. This heat transfer occurs via conduction through the building envelope (walls, fenestrations, and ceiling) as a result of ambient outdoor temperatures and the solar heat absorbed by the building's exterior surfaces.

- **Heat Transmission Through Glass Windows and Doors**

The doors and windows on the front and rear facades of the laboratory contribute to the overall heat load through both thermal conduction and diffuse solar radiation. Accounting for these specific heat gains is essential when determining the required cooling capacity of the new air conditioning system. The following assumptions were made to calculate the heat transmission through these elements.

- Outdoor temperature = 32 °C
- Indoor temperature = 24 °C
- Windows and Door (type 3) are made of single clean glass with 70% UV protection.
- Heat transfer due to convection is considered negligible.
- Solar heat gain factor (SHGF) for in Sri Lanka is assumed as 500Wm⁻²
- Convective heat loss through windows is negligible.

- **Heat Conduction Through Walls**

Thermal conduction through the laboratory walls constitutes a significant portion of the total cooling load. This heat transfer is driven by the temperature gradient across the structural partitions—specifically the difference between the conditioned interior space and the adjacent unconditioned rooms or outdoor environment. To accurately determine the specific thermal boundary conditions for each wall segment, the surrounding architectural layout of the Computer Laboratory is illustrated in the figure below.

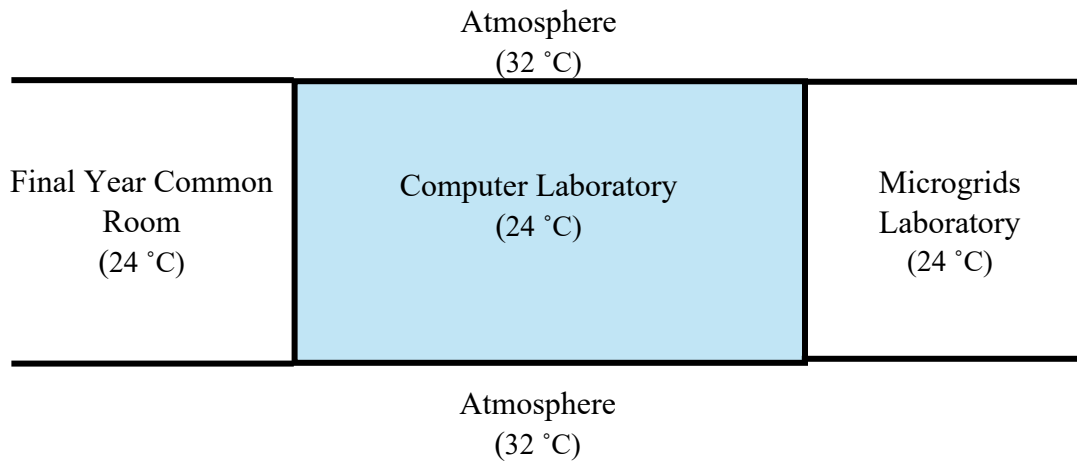


Figure 05 – Surrounding arrangement of the laboratory

The wall of the laboratory has the following structure.

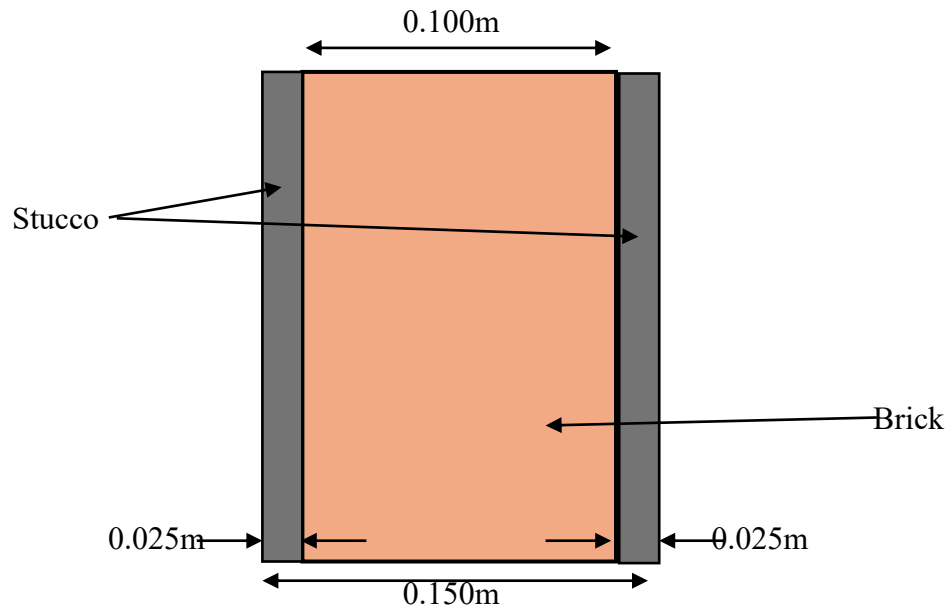


Figure 06 – Wall structure of the laboratory

Material	Thickness (d)	Thermal Conductivity (c)	Thermal Resistance (d/c)
Brick	0.100m	0.89 W/mK	0.1124 m ² K/W
Stucco	0.025m	0.72 W/mK	0.0347 m ² K/W

Overall heat transfer coefficient (U) can be calculated as follows.

$$\text{Total resistance } (R_{eq}) = 0.1124 + 2 \times 0.0347 = 0.1818 \text{ m}^2 \text{K/W}$$

$$U_1 = 1/R_{eq} = 0/0.1818$$

$$U_1 = 5.501 \text{ W/m}^2 \text{K}$$

The following assumptions were made when calculating the heat conduction through walls.

- The Computer Laboratory, Microgrids Laboratory, and Final Year Common Room are all assumed to operate at a continuous internal temperature of 24 °C. Therefore, heat transfer through the internal walls shared with these spaces is zero.
- The ambient outdoor design temperature is assumed to be 32 °C.
- Heat conduction through the slab is negligible.

• Heat Conduction Through Wooden Doors

When calculating the heat load, the heat conduction through the two wooden doors in the laboratory should also be considered. Following assumptions were made when calculating the heat conduction through wooden doors.

- Doors are assumed as 0.05m thickness wooden doors.

Heat transfer coefficient for the doors can be calculated as follows.

Material	Thickness (d)	Thermal Conductivity (c)	Thermal Resistance (d/c)
Wood	0.050m	0.15 W/mK	0.333 m ² K/W

$$\text{Total resistance } (R_{eq}) = 0.333 \text{ m}^2 \text{K/W}$$

$$U_2 = 1/R_{eq} = 0/0.333$$

$$U_2 = 3 \text{ W/m}^2 \text{K}$$

- **Air Infiltration**

Another critical external heat source is infiltration, which accounts for the heat gain caused by warm outdoor air seeping into the cooled laboratory through unintended openings around the doors and windows.

➤ Air change per hour (ACH) was assumed as 1 when calculating the air infiltration.

04. Heat Load Calculation

Heat Source		Calculation
Internal Heat Sources	Heat gain by desktop computers ($Q_{computer}$)	$= N \times \text{Peak heat gain}$ $= 70 \times 59$ $= 4130W$
	Heat gains from people (Q_{people})	$= N \times (SGH + LHG)$ $= 60 \times (75 + 55)$ $= 6500W$
	Heat gain of lights (Q_{light})	$= W_{total} \times F_{ul} \times F_{sa}$ $= 34 \times 22 \times 0.9 \times 1$ $= 673.2W$
External Heat Sources	Heat gain through glass windows and doors (Q_{glass})	$= Q_{solar} + Q_{conductive} + Q_{convective}$ $= (A \times SHGF \times SC) + (UA\Delta\theta) + 0$ $= (34.718 \times 500 \times (1 - 0.7))$ $\quad + 5.8 \times 34.718 \times (32 - 24)$ $= 6818.62W$
	Heat conduction through walls (Q_{wall})	$= Q_{wall1} + Q_{wall2}$ $= U_1A_1\Delta\theta + U_1A_2\Delta\theta$ $= 5.501 \times 35.89 \times (32 - 24)$ $\quad + 5.501 \times 41.27 \times (32 - 24)$ $= 3395.66W$
	Heat conduction through wooden doors (Q_{door})	$= Q_{door1} + Q_{door2}$ $= U_2A_1\Delta\theta + U_1A_2\Delta\theta$ $= 3 \times 3.36 \times (32 - 24) + 3 \times 2.15 \times (32 - 24)$ $= 132.24W$
	Air infiltration ($Q_{infiltration}$)	$= 0.33 \times V \times ACH \times \Delta\theta$ $= 0.33 \times 560.297 \times 1 \times (32 - 24)$ $= 1479.18W$

*All the values for the calculations were obtained using the tables in Appendix 1.

Total cooling load can be calculated as follows.

$$Q_{total} = Q_{computer} + Q_{people} + Q_{light} + Q_{glass} + Q_{wall} + Q_{door} + Q_{infiltration}$$

$$Q_{total} = 4130 + 6500 + 673.2 + 6818.62 + 3395.66 + 132.24 + 1479.18$$

$$Q_{total} = 23128.9W$$

$$Q_{total} = 3.41 \times 23128.9 \text{ BTU/h}$$

$$\underline{Q_{total} = 78869.549 \text{ BTU/h}}$$

05. Selection of Air Conditioning System

Based on the comprehensive thermal analysis, the calculated peak heat load for the Computer Laboratory is 78,869 BTU/h. To effectively manage this thermal demand and account for real-world performance degradation, such as dust accumulation and extreme ambient temperature spikes, a standard engineering safety margin of 10% to 20% is recommended. Therefore, a practical and highly efficient solution is the installation of two commercial inverter type units, each with a capacity of 48,000 BTU/h. This configuration provides a total cooling capacity of 96,000 BTU/h, safely satisfying the requirement while ensuring system longevity.

05.1 Cooling Capacity Requirements and Selection

- Calculated Heat Load: 78,869 BTU/h
- Proposed System: 2 × 48,000 BTU/h Inverter-Type units
- Total Installed Capacity: 96,000 BTU/h (> 78,869 BTU/h)

05.2 Justification for the Proposed Configuration

➤ Improved Air Distribution

The laboratory is a large rectangular space accommodating 70 students and their equipment. Utilizing two distinct 48,000 BTU/h units, positioned strategically at opposite ends of the room, prevents localized hot spots, ensures a uniform temperature gradient, and reduces the thermal strain on a single localized compressor.

➤ Energy Efficiency (Inverter Technology)

Given that the laboratory functions as both a teaching space and a study area, it is expected to be air-conditioned for extended periods (approximately 8–12 hours per day). Inverter technology is critical here; it modulates the compressor speed to maintain the 24°C setpoint continuously, eliminating the high power spikes of traditional non-inverter systems and significantly reducing the department's operational energy footprint.

➤ Installation Considerations (Ceiling-Suspended Units)

To minimize complex ductwork installation and accommodate the structural constraints of the Sumanadasa building's concrete slab, ceiling-suspended (under-ceiling) air conditioners are preferred. These units are highly suitable because they offer following specifications.

- Do not require deep false-ceiling clearances for installation.
- Deliver powerful, long-throw horizontal airflow that can easily cover the expansive length of the 70 student laboratory.
- Operate quietly, maintaining an acoustic environment suitable for academic instruction.

Specific Market Availability & Final Recommendation

Install **2 × Ceiling-Suspended Inverter-Type Air Conditioners (48,000 BTU/h each)**.

Based on current market availability in Sri Lanka, a highly suitable model for this application is the **Singer Air Conditioner Ceiling Mounted Inverter 48000 BTU (Model: SASI48INVCN)**. Alternatively, the Gree Ceiling Suspended 48000 BTU Inverter AC is also a viable commercial option.

06. Heat Load Calculation Through Online Software

In addition to the manual heat load calculations, a software-based assessment was performed to provide a point of comparison. The online "Manual J Load Calculator" (<https://share.google/Z5uCT9dREdxHjtqgV>) was utilized for this purpose. The following analysis details the software's load estimation.

Manual J Load Calculation

1 · Building	2 · Envelope	3 · Openings	4 · Loads & Temps
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Building Basics

Conditioned Floor Area (sq ft)

1723.14

Ceiling Height (ft)

11.48

Higher ceilings increase the air volume requiring conditioning

Climate Zone

Mixed Humid (GA, NC, TN, VA, MO, KS)

Building Orientation

Mainly East-facing

South and west-facing buildings have higher cooling loads from solar gain

Manual J Load Calculation

1 · Building

2 · Envelope

3 · Openings

4 · Loads & Temps

Envelope and Insulation

Insulation Quality

Average (code minimum, R-13 walls)



Air Tightness

Average (typical construction)



Sun Exposure

Average (typical suburban lot)



Window Type

Single-pane (U-1.0+)



Manual J Load Calculation

1 · Building

2 · Envelope

3 · Openings

4 · Loads & Temps

Windows and Doors

Number of Windows

22

Exterior Doors

2

Assumes approx. 15 sq ft each. Count all exterior-facing windows.

Door Type

Standard wood



Attached Conditioned Sunroom

No sunroom



Manual J Load Calculation

1 · Building

2 · Envelope

3 · Openings

4 · Loads & Temps

Internal Loads and Design Temperatures

Occupants

60

Each person adds ~250 BTU/hr cooling load and ~100 BTU/hr heating load

Kitchen Appliance Load

No kitchen

Major Appliances (count)

70

Refrigerators, dishwashers, dryers, etc.

Total Lighting Wattage (W)

748

1 watt = 3.41 BTU/hr cooling load

Summer Outdoor Design Temp (°F)

89.6

ACCA 99% design temp for your city - not the hottest day on record

Winter Outdoor Design Temp (°F)

89.6

ACCA 1% design temp for your city - use ACCA Manual J design data tables

Indoor Design Temperature (°F)

74.2

Disclaimer: This is a simplified Manual J load calculation estimate using adjusted square footage methodology. Results are typically within 10-15% of full Manual J software for standard residential homes. For code-required permit documentation, use ACCA-approved software (Wrightsoft, CoolCalc, Elite RHVAC). Not a substitute for a licensed HVAC engineer's analysis on complex or unusual structures.

Manual J Load Calculation Results

HEATING LOAD

24,800

BTU/hr - 2 tons

COOLING LOAD

85,400

BTU/hr - 7 tons

RECOMMENDED

7 ton

cooling-dominant

SENSIBLE / LATENT

62,342

/ 23,058 BTU (est.)

Manual J estimate for 1,723.14 sq ft mixed humid climate building: heating load 24,800 BTU (2 ton) / cooling load 85,400 BTU (7 ton). Recommended starting size: 7 tons (cooling-dominant). Results pre-fill the Manual S equipment selection calculator.



Large load detected (5+ tons). Consider zoned system or multiple units. Full room-by-room Manual J from ACCA-approved software is strongly recommended at this scale.

Appendix 1

Data Sheets

Table 1 Representative Rates at Which Heat and Moisture Are Given Off by Human Beings in Different States of Activity

Degree of Activity	Location	Total Heat, W		Sensible Heat, W	Latent Heat, W	% Sensible Heat that is Radiant ^b	
		Adult Male	Adjusted, M/F ^a			Low <i>V</i>	High <i>V</i>
Seated at theater	Theater	115	105	70	35	60	27
Seated, very light work	Offices, hotels, apartments	130	115	70	45		
Moderately active office work	Offices, hotels, apartments	140	130	75	55		
Standing, light work; walking	Department store; retail store	160	130	75	55	58	38
Walking, standing	Drug store, bank	160	145	75	70		
Sedentary work	Restaurant ^c	145	160	80	80		
Light bench work	Factory	235	220	80	140		
Moderate dancing	Dance hall	265	250	90	160	49	35
Walking 4.8 km/h; light machine work	Factory	295	295	110	185		
Bowling ^d	Bowling alley	440	425	170	255		
Heavy work	Factory	440	425	170	255	54	19
Heavy machine work; lifting	Factory	470	470	185	285		
Athletics	Gymnasium	585	525	210	315		

Notes:

1. Tabulated values are based on 24°C room dry-bulb temperature. For 27°C room dry bulb, total heat remains the same, but sensible heat values should be decreased by approximately 20%, and latent heat values increased accordingly.

2. Also see Table 4, Chapter 9, for additional rates of metabolic heat generation.

3. All values are rounded to nearest 5 W.

^aAdjusted heat gain is based on normal percentage of men, women, and children for the application listed, and assumes that gain from an adult female is 85% of that for an adult male, and gain from a child is 75% of that for an adult male.

^bValues approximated from data in Table 6, Chapter 9, where *V* is air velocity with limits shown in that table.

^cAdjusted heat gain includes 18 W for food per individual (9 W sensible and 9 W latent).

^dFigure one person per alley actually bowling, and all others as sitting (117 W) or standing or walking slowly (231 W).

Table 2 Lighting Power Densities Using Space-by-Space Method

Common Space Types ^a	LPD, W/m ²	Common Space Types ^a	LPD, W/m ²	Building-Specific Space Types ^a	LPD, W/m ²
Atrium		Loading Dock, Interior	5.1	Health Care Facility	
≤12.2 m high	1.1/m total height	Lobby		In exam/treatment room	18.0
>12.2 m high	4.3 + 0.7/m total height	In facility for the visually impaired (and not used primarily by staff) ^c	19.4	In imaging room	16.3
		For elevator	7.0	In medical supply room	
Audience Seating Area		In hotel	11.5	In nursery	9.5
In auditorium	6.8	In motion picture theater	6.4	In nurses' station	7.6
In convention center	8.9	In performing arts theater	21.6	In operating room	26.8
In gymnasium	7.1	All other lobbies	9.7	In patient room	6.7
In motion picture theater	12.3	Locker Room	8.1	In physical therapy room	9.9
In penitentiary	3.1	Lounge/Breakroom		In recovery room	12.4
In performing arts theater	26.2	In health care facility	10.0	Library	
In religious building	16.5	All other lounges/breakrooms	7.9	In reading area	11.5
In sports arena	4.7	Office		In stacks	18.4
All other audience seating areas	4.7	Enclosed	12.0	Manufacturing Facility	
Banking Activity Area	11.9	Open plan	10.6	In detailed manufacturing area	13.9
Breakroom (See Lounge/Breakroom)		Parking Area, Interior	2.1	In equipment room	8.0
Classroom/Lecture Hall/Training Room		Pharmacy Area	18.1	In extra-high-bay area (15.2 m floor-to-ceiling height)	11.3
In penitentiary	14.5	Restroom		In high-bay area (7.6 to 15.2 m floor-to-ceiling height)	13.3
All other classrooms/lecture halls/training rooms	13.4	In facility for the visually impaired (and not used primarily by staff) ^c	13.1	In low-bay area (<7.6 m floor-to-ceiling height)	12.9
Conference/Meeting/Multipurpose Room	13.3	All other restrooms	10.6	Museum	
Confinement Cells	8.8	Sales Area^d	15.5	In general exhibition area	11.4
Copy/Print Room	7.8	Seating Area, General	5.9	In restoration room	11.0
Corridor^b		Stairway		Performing Arts Theater, Dressing Room	6.6
In facility for visually impaired (and not used primarily by staff) ^c	9.9	Space containing stairway determines LPD and control requirements for stairway.		Post Office, Sorting Area	10.2
In hospital	10.7	Stairwell	7.4	Religious Buildings	
In manufacturing facility	4.4	Storage Room		In fellowship hall	6.9
All other corridors	7.1	<4.65 m ²	13.3	In worship/pulpit/choir area	16.5
Courtroom	18.6	All other storage rooms	6.8	Retail Facilities	
Computer Room	18.4	Vehicular Maintenance Area	7.3	In dressing/fitting room	7.7
Dining Area		Workshop	17.2	In mall concourse	11.9
In penitentiary	10.4	Building-Specific Space Types^a	LPD, W/m²	Sports Arena, Playing Area	
In facility for visually impaired (and not used primarily by staff) ^c	28.5 W/m ²	Facility for Visually Impaired^e		For Class I facility	39.7
In bar/lounge or leisure dining	11.6	In chapel (used primarily by residents)	23.8	For Class II facility	25.9
In cafeteria or fast food dining	7.0	In recreation room/common living room (and not used primarily by staff)	26.0	For Class III facility	19.4
In family dining	9.6	Automotive (See Vehicular Maintenance Area)		For Class IV facility	13.0
All other dining areas	7.0	Convention Center: Exhibit Space	15.7	Transportation Facility	
Electrical/Mechanical Room^f	4.6	Dormitory/Living Quarters	4.2	In baggage/carousel area	5.7
Emergency Vehicle Garage	6.1	Fire Station: Sleeping Quarters	0.22	In an airport concourse	3.9
Food Preparation Area	13.1	Gymnasium/Fitness Center		At a terminal ticket counter	8.7
Guest Room	9.8	In exercise area	7.8	Warehouse—Storage Area	
Laboratory		In playing area	13.0	For medium to bulky, palletized items	6.2
In or as classroom	15.5			For smaller, hand-carried items ^g	10.2
All other laboratories	19.5				
Laundry/Washing Area	6.5				

Source: ASHRAE Standard 90.1-2013.

^aIn cases where both a common space type and a building-specific type are listed, the building-specific space type applies.

^cA facility for the visually impaired one that can be documented as being designed to comply with light levels in ANSI/IES RP-28 and is (or will be) licensed by local/state authorities for either senior long-term care, adult

^dFor accent lighting, see section 9.6.2(b) of ASHRAE Standard 90.1-2013.

^eSometimes called a picking area.

^fAn additional 5.7 W/m² is allowed only if this additional

Table 8B Recommended Heat Gain for Typical Laptops and Laptop Docking Station

Equipment Description		Name-plate Power, ^a W	Peak Heat Gain, ^{b, c} W
Laptop computer	Manufacturer 1, 2.6 GHz processor, 8 GB RAM, $n = 1$	NA	46
	Manufacturer 2, 2.4 GHz processor, 4 GB RAM, $n = 1$	NA	59
Average 15-min peak power consumption (range)		53 (46-59)	
Laptop with docking station	Manufacturer 1, 2.7 GHz processor, 8 GB RAM, $n = 1$	NA	38
	1.6 GHz processor, 8 GB RAM, $n = 2$	NA	45
	2.0 GHz processor, 8 GB RAM, $n = 1$	NA	50
	2.6 GHz processor, 4 GB RAM, $n = 1$	NA	51
	2.4 GHz processor, 8 GB RAM, $n = 1$	NA	40
	2.6 GHz processor, 8 GB RAM, $n = 1$	NA	35
	2.7 GHz processor, 8 GB RAM, $n = 1$	NA	59
	3.0 GHz processor, 8 GB RAM, $n = 3$	NA	70
	2.9 GHz processor, 32 GB RAM, $n = 3$	NA	58
	3.0 GHz processor, 32 GB RAM, $n = 1$	NA	128
	3.7 GHz processor, 32 GB RAM, $n = 1$	NA	63
	3.1 GHz processor, 32 GB RAM, $n = 1$	NA	89
Average 15-min peak power consumption (range)		61 (26-151)	

Source: Bach and Sarfraz (2017)

n = number of tested equipment of same configuration.

^aVoltage and amperage information for laptop computer and laptop docking station is available on power supply nameplates; however, nameplate does not provide information on power consumption, where NA = not available.

^bFor equipment peak heat gain value, the highest 15-min interval of recorded data is listed in tables.

^cApproximately 75% convective heat gain and 25% radiative heat gain.

Table 9 Recommended Heat Gain for Typical Printers

Equipment	Description	Max. Printing Speed, Pages per Minute	Name- plate Power, W	Peak Heat Gain,^a W
Multifunction printer (copy, print, scan)	Large, multiuser, office type	40	1010	540 (Idle 29 W)
		30	1300	303 (Idle 116 W)
		28	1500	433 (Idle 28 W)
	Average 15-min peak power consumption (range)		425 (303-540)	
	Multiuser, medium-office type	35	900	732 (Idle 18 W)
	Desktop, small-office type	25	470	56 (Idle 3 W)
Monochrome printer	Desktop, medium-office type	55	1000	222
		45	680	61
	Average 15-min peak power consumption (range)		142 (61-222)	
Color printer	Desktop, medium-office type	40	620	120
Laser printer	Desktop, small-office type	14	310	89
		24	495	67
		26	1090	65
	Average 15-min peak power consumption (range)		74 (65-89)	
Plotter	Manufacturer 1		1600	571
			270	173
	Average 15-min peak power consumption (range)		372 (173-571)	
Fax machine	Medium		1090	92
	Small		600	46
	Average 15-min peak power consumption (range)		69 (46-92)	

Source: Bach and Sarfraz (2017)

^aApproximately 70% convective heat gain and 30% radiative heat gain.

Table 18 Thermal Properties and Code Numbers of Layers Used in Wall and Roof Descriptions for Tables 16 and 17

Layer ID	Description	Thickness, mm	Conductivity, W/(m·K)	Density, kg/m ³	Specific Heat, kJ/(kg·K)	Resistance R, (m ² ·K)/W	R	Mass, kg/m ²	Thermal Capacity, kJ/(m ² ·K)	Notes
F01	Outdoor surface resistance	—	—	—	—	0.04	0.04	—	—	1
F02	Indoor vertical surface resistance	—	—	—	—	0.12	0.12	—	—	2
F03	Indoor horizontal surface resistance	—	—	—	—	0.16	0.16	—	—	3
F04	Wall air space resistance	—	—	—	—	0.15	0.15	—	—	4
F05	Ceiling air space resistance	—	—	—	—	0.18	0.18	—	—	5
F06	EIFS finish	9.5	0.72	1856	0.84	—	0.01	17.7	14.92	6
F07	25 mm stucco	25.4	0.72	1856	0.84	—	0.04	47.2	39.45	6
F08	Metal surface	0.8	45.28	7824	0.50	—	0.00	6.0	3.07	7
F09	Opaque spandrel glass	6.4	0.99	2528	0.88	—	0.01	16.1	14.10	8
F10	25 mm stone	25.4	3.17	2560	0.79	—	0.01	65.1	51.71	9
F11	Wood siding	12.7	0.09	592	1.17	—	0.14	7.5	8.79	10
F12	Asphalt shingles	3.2	0.04	1120	1.26	—	0.08	3.6	4.50	
F13	Built-up roofing	9.5	0.16	1120	1.46	—	0.06	10.7	15.74	
F14	Slate or tile	12.7	1.59	1920	1.26	—	0.01	24.4	30.67	
F15	Wood shingles	6.4	0.04	592	1.30	—	0.17	3.8	4.91	
F16	Acoustic tile	19.1	0.06	368	0.59	—	0.31	7.0	4.09	11
F17	Carpet	12.7	0.06	288	1.38	—	0.22	3.7	5.11	12
F18	Terrazzo	25.4	1.80	2560	0.79	—	0.01	65.1	51.71	13
G01	16 mm gyp board	15.9	0.16	800	1.09	—	0.10	12.7	13.90	
G02	16 mm plywood	15.9	0.12	544	1.21	—	0.14	8.6	10.42	
G03	13 mm fiberboard sheathing	12.7	0.07	400	1.30	—	0.19	5.1	6.54	14
G04	13 mm wood	12.7	0.15	608	1.63	—	0.08	7.7	12.67	15
G05	25 mm wood	25.4	0.15	608	1.63	—	0.17	15.5	25.35	15
G06	50 mm wood	50.8	0.15	608	1.63	—	0.33	30.9	50.49	15
G07	100 mm wood	101.6	0.15	608	1.63	—	0.66	61.8	100.97	15
I01	25 mm insulation board	25.4	0.03	43	1.21	—	0.88	1.1	1.43	16
I02	50 mm insulation board	50.8	0.03	43	1.21	—	1.76	2.2	2.66	16
I03	75 mm insulation board	76.2	0.03	43	1.21	—	2.64	3.3	4.09	16
I04	89 mm batt insulation	89.4	0.05	19	0.96	—	1.94	1.7	1.64	17
I05	154 mm batt insulation	154.4	0.05	19	0.96	—	3.34	3.0	2.86	17
I06	244 mm batt insulation	243.8	0.05	19	0.96	—	5.28	4.7	4.50	17
M01	100 mm brick	101.6	0.89	1920	0.79	—	0.11	195.2	155.34	18
M02	150 mm LW concrete block	152.4	0.49	512	0.88	—	0.31	78.1	68.68	19
M03	200 mm LW concrete block	203.2	0.50	464	0.88	—	0.41	94.3	82.99	20
M04	300 mm LW concrete block	304.8	0.71	512	0.88	—	0.43	156.2	137.36	21
M05	200 mm concrete block	203.2	1.11	800	0.92	—	0.18	162.7	149.83	22
M06	300 mm concrete block	304.8	1.40	800	0.92	—	0.22	244.0	224.84	23
M07	150 mm LW concrete block (filled)	152.4	0.29	512	0.88	—	0.53	78.1	68.68	24
M08	200 mm LW concrete block (filled)	203.2	0.26	464	0.88	—	0.78	94.3	82.99	25
M09	300 mm LW concrete block (filled)	304.8	0.29	512	0.88	—	1.04	156.2	137.36	26
M10	200 mm concrete block (filled)	203.2	0.72	800	0.92	—	0.28	162.7	149.83	27
M11	100 mm lightweight concrete	101.6	0.53	1280	0.84	—	0.19	130.1	108.95	
M12	150 mm lightweight concrete	152.4	0.53	1280	0.84	—	0.29	195.2	163.52	
M13	200 mm lightweight concrete	203.2	0.53	1280	0.84	—	0.38	260.3	218.10	
M14	150 mm heavyweight concrete	152.4	1.95	2240	0.90	—	0.08	341.6	307.62	
M15	200 mm heavyweight concrete	203.2	1.95	2240	0.90	—	0.10	455.5	410.23	
M16	300 mm heavyweight concrete	304.8	1.95	2240	0.90	—	0.16	683.2	615.24	
M17	50 mm LW concrete roof ballast	50.8	0.19	840	0.84	—	0.27	32.5	27.19	28